
EE 708 : Information Theory and Coding

IIT Bombay, Dept. of EE

Instructor: B. K. Dey

Numerical (and a little analytical) homework on AEP

Notation: $H_2(x) := -x \log_2(x) - (1-x) \log_2(1-x)$

and $D_2(x||y) := x \log_2 \frac{x}{y} + (1-x) \log_2 \frac{1-x}{1-y}$

Question: 1 Consider a binary random variable $X \sim \text{Bern}(p)$ for $p = 0.4$, and suppose $n = 10$.

(a) The number of n -length binary sequences having $n\kappa$ ones is given by $f_n(\kappa) = \binom{n}{n\kappa}$. Plot this function against κ , keeping n fixed and varying κ from 0 to 1 in steps of $1/n$.

(b) The probability of an n -length sequence with $n\kappa$ ones is given by $g_n(\kappa) = p^{n\kappa}(1-p)^{n(1-\kappa)}$. In the same figure, plot the functions $g_n(\kappa)$ against κ . In another figure, plot $-\frac{1}{n} \log_2 g_n(\kappa)$ and $D_2(\kappa||p) + H_2(\kappa)$ against κ and compare the two plots. Observe how the difference between them changes when you take $n = 20$.

(c) The probability of observing $n\kappa$ ones is given by $\binom{n}{n\kappa} p^{n\kappa} (1-p)^{n(1-\kappa)} = f_n(\kappa) g_n(\kappa)$. Note that the fraction K of observed ones is a random variable, and $f_n(\kappa) g_n(\kappa)$ is its probability mass function. Plot $-\frac{1}{n} \log_2 f_n(\kappa) g_n(\kappa)$ against κ . In the same figure, plot $D_2(\kappa||p)$ and compare the two plots. Observe how the difference between them changes when you take $n = 20$.

(d) Both for $n = 10$, $n = 15$ and $n = 20$, take $\epsilon = 0.2$, and compute $\Pr(p - \epsilon \leq K \leq p + \epsilon)$. Do you see the higher concentration of K reflected as n increases?

(e) Now note that the probability of each of these sequences (with $\sim np$ ones each) is given by (approximately)

Correlate the above observations with theory discussed in class. In particular, recall that for a vector of length n drawn i.i.d. $\sim p$,

Number of sequences of type $q \approx 2^{nH(q)}$

Probability of a type q sequence $\approx 2^{-n(D(q||p)+H(q))}$.

Probability of the type class of type q is $\approx 2^{-nD(q||p)}$.